

Sample Main Alerts + Sub Alerts Data (5 Sets)

1. Database Outage

Main Alert Title:

DB-01 Primary Database Connection Failure

Sub Alerts (12)

1. ERROR: Connection timeout to db-01:5432 from payment-api
2. CRITICAL: pg_isready failed on db-01 after 30s
3. WARNING: Connection pool exhausted for service order-api targeting db-01
4. ERROR: FATAL: could not connect to server db-01 from checkout-service
5. ALERT: Database connection refused from inventory-api to db-01
6. ERROR: timeout expired when connecting to PostgreSQL db-01
7. CRITICAL: Health check failed: db-01 not responding
8. ERROR: Npgsql.NpgsqlException: Exception while connecting to db-01
9. WARNING: Slow query detected, possible db-01 saturation
10. ERROR: Failed to acquire connection from pool - db-01
11. ALERT: Replication lag critical on db-01 replica
12. ERROR: 500 Internal Server Error - database unavailable db-01

Common Theme: All alerts point to **db-01** database connectivity or availability issues.

2. Payment Gateway Down

Main Alert Title:

Stripe Payment Gateway 5xx Error Spike

Sub Alerts (10)

1. ERROR: Stripe API returned 503 Service Unavailable from payment-service
2. ALERT: Payment processing failed: Gateway timeout stripe.com
3. CRITICAL: 100% payment failures in last 5 min - Stripe
4. ERROR: Webhook delivery failed from Stripe - 502 Bad Gateway
5. WARNING: Increased latency to api.stripe.com - 8s response

6. ERROR: StripeException: Connection to Stripe API failed
7. ALERT: Checkout blocked - cannot reach payment gateway
8. ERROR: Refund API failing - Stripe 504 Gateway Timeout
9. CRITICAL: Payment service circuit breaker OPEN for Stripe
10. ERROR: Unable to create Stripe customer - service down

Common Theme: All alerts indicate **Stripe payment gateway outage or degradation**.

3. High CPU on Kubernetes Nodes

Main Alert Title:

K8s Worker Node Pool CPU Saturation

Sub Alerts (15)

1. WARNING: CPU usage 95% on k8s-worker-03 for 10m
2. CRITICAL: Node k8s-worker-03 NotReady - CPU pressure
3. ALERT: Pod evictions started on k8s-worker-03 due to CPU
4. WARNING: CPU throttling on namespace prod, node k8s-worker-03
5. ERROR: Failed to schedule pods - insufficient CPU k8s-worker-03
6. ALERT: Load average 24.5 on k8s-worker-03, expected <8
7. WARNING: CPU usage 92% on k8s-worker-05
8. CRITICAL: Node k8s-worker-05 memory + CPU pressure
9. ALERT: HorizontalPodAutoscaler failed - no CPU available
10. WARNING: Container restart count high on k8s-worker-03 - OOMKill
11. ERROR: Liveness probe failed due to CPU starvation node-03
12. ALERT: kubelet CPU usage 80% on k8s-worker-03
13. WARNING: Node k8s-worker-04 CPU 88% sustained
14. CRITICAL: Cluster CPU utilization 85% - scale up needed
15. ALERT: Prometheus alert: NodeCPUHigh on k8s-worker-03

Common Theme: Multiple Kubernetes worker nodes experiencing **CPU saturation and resource pressure**.

4. Auth Service Crash Loop

Main Alert Title:

Auth Microservice CrashLoopBackOff in Production

Sub Alerts (11)

1. ERROR: Pod auth-service-7d8f9 crashed - Exit Code 1
2. WARNING: auth-service restarting - CrashLoopBackOff detected
3. CRITICAL: auth-service deployment has 0/3 pods ready
4. ERROR: Panic in auth-service: nil pointer dereference
5. ALERT: Failed to pull image auth-service:v2.4.1
6. ERROR: Liveness probe failed auth-service port 8080
7. WARNING: Readiness probe failed - /health returning 500
8. ALERT: OOMKilled: auth-service exceeded 512Mi limit
9. ERROR: ConfigMap auth-config not found - pod crash
10. CRITICAL: All login attempts failing - auth service down
11. ERROR: JWT validation service unreachable - auth pod down

Common Theme: The **auth-service** is repeatedly crashing, causing authentication failures.

5. AWS Availability Zone Network Partition

Main Alert Title:

AWS us-east-1a Availability Zone Network Issue

Sub Alerts (13)

1. ALERT: Packet loss 40% from instances in us-east-1a
2. ERROR: EC2 instance i-0abc123 unreachable - us-east-1a
3. WARNING: High network latency us-east-1a to us-east-1b - 300ms
4. CRITICAL: ELB health checks failing for 1a targets
5. ERROR: RDS connection timeout - endpoint in us-east-1a
6. ALERT: S3 API errors from us-east-1a - 503 Slow Down
7. WARNING: VPN tunnel down to us-east-1a VPC
8. ERROR: DNS resolution failed for internal services in 1a

- 9. ALERT: NAT Gateway us-east-1a degraded performance
- 10. CRITICAL: Cross-AZ traffic from 1a dropping packets
- 11. ERROR: EKS node group 1a nodes NotReady
- 12. WARNING: CloudWatch metrics delayed from us-east-1a
- 13. ALERT: Intermittent connectivity to us-east-1a resources

Common Theme: Widespread network degradation affecting resources in **AWS us-east-1a Availability Zone**.

Summary Table

Main Alert	Sub Alerts	Key Correlation Keyword
DB-01 Primary Database Connection Failure	12	db-01
Stripe Payment Gateway 5xx Error Spike	10	Stripe
K8s Worker Node Pool CPU Saturation	15	CPU, k8s-worker-*
Auth Microservice CrashLoopBackOff in Production	11	auth-service
AWS us-east-1a Availability Zone Network Issue	13	us-east-1a

Total Alerts: 61
Expected Main Incidents after Correlation: 5
Noise Reduction: ~91.8% (61 raw alerts → 5 grouped incidents)